

Phase 1 Poster Content — 7 October 2026

Poster session focus (course brief): "compilation of usability studies and design themes for your project". Suggested layout: A1 portrait, 3 columns × 3 rows. Each block below is one panel. Keep text ≤ 60 words per panel; use the figures listed.

Panel 1 — Title

Campus Entry Authentication: from a glance at a card to authenticated, authorized, private entry CSE 347/652 | DES 306/525 Usable Security and Privacy · IIIT-Delhi · Monsoon 2026 Team: [INSERT NAMES]

Panel 2 — The problem (with current-workflow sketch, Section 8)

A guard glances at a printed ID. The card proves possession, not identity. Lost, borrowed, forged, expired and revoked cards look the same. No revocation check. No audit trail. Visitors' phone numbers in a public register. At 8:55 a.m., the check becomes a wave.

Panel 3 — How we studied it (Section 9)

Three primary methods + secondary research:

- Student questionnaire — 20 items: frequency, forgotten/lost ID, wait, security & privacy perceptions, method preference, accessibility
- Guard interviews — 14 questions: real workflow, suspicious/fake/lost IDs, visitors, peak load, what they need to see and what they should not
- Contextual inquiry — timed observation, U/Q/N/V/R coding, no identifying data

Status: [INSERT n students, n guards, n sessions — or "fieldwork in progress; no data fabricated"]

Panel 4 — What we learned

[DATA REQUIRED] Replace this panel with 3–5 real findings and one anonymised quote each. Until then, show the hypotheses H1–H9 and label them "to be validated". Secondary evidence we can already state: MIFARE Classic and hotel-lock systems covering millions of doors were broken by cloning data from a single card [4, 5, 8, 9]; users and operators bypass slow security rationally [10, 11, 17].

Panel 5 — Design themes (Section 33)

1. Authenticate the key, not the card — challenge–response; never accept a bare UID or static QR
2. Two questions, two answers — "Who are you?" and "Are you allowed here, now?" fail differently and say different things
3. Fail secure, degrade gracefully — offline-verified ≤4 h → supervised manual mode; never an open gate, never "allow anyway"
4. Show the guard a face and a status, nothing more — photo from the server, first name, role, decision
5. Dignity on denial — generic message to bystanders, private explanation to the student
6. Accountability without surveillance — pseudonymous logs, justified re-identification, 90-day retention

Panel 6 — QR vs RFID vs NFC (Section 13 table, condensed)

	Static QR	LF RFID	AES smart card	NFC phone	Dynamic QR
Cloning	×	×	✓	✓	•
Replay	×	×	✓	✓	•
Speed	•	✓	✓	✓	•
Works for everyone	✓	✓	✓	×	×

Provisional direction: AES smart card for all + optional phone credential + signed visitor passes. Pending Q16/Q17 data.

Panel 7 — Threat model (Figure J-9 attack tree)

12 actors · 17 threats · 10 scenarios. Highest priority: theft, cloning, replay, impersonation, revoked reuse, privilege escalation, DB compromise, rogue server, privacy leakage, log tampering, reader compromise, insider abuse. Residual, stated plainly: relay attacks (mitigated only by the human face check), guard fatigue, offline staleness window.

Panel 8 — Prototype (show S05, S06, S08, S12 wireframes)

20 lo-fi screens · 3 surfaces (gate indicator, guard tablet, admin/student). Every DENY screen ends with a next step. No guard screen has an "Allow" button. Guard sees photo + first name + role only. Initial heuristic evaluation (not user testing): 15 findings, 3 major, all about denial wording → fixed in next iteration.

Panel 9 — Traceability & next steps (Section 37, Figure J-2)

Finding → need → requirement → threat → feature → screen → test. 20 FR · 20 SEC · 12 PRIV · 14 USE · 14 NFR, each mapped to a screen and a Phase 2 test. Phase 2: complete fieldwork → implement with standard crypto only (Ed25519, AES, TLS 1.3) → attack-scenario tests → guard & student usability tests → executable + user guide 17 Nov.